

Business Requirements Document (BRD)

Project Title: Customer Analytics Platform for Strategic Customer Retention and Revenue Growth

Dataset: Olist Brazilian E-commerce Dataset

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1. Executive Summary

The rapid expansion of e-commerce has intensified competition, increased customer acquisition costs, and raised customer expectations regarding service quality and personalization. While organizations collect vast amounts of transactional and customer data, many struggle to convert this information into actionable insights that improve business performance.

This project aims to develop a Customer Analytics Platform that enables data-driven decision-making across customer relationship management, marketing, operations, and executive leadership. Using transactional data from the Olist Brazilian E-Commerce Public Dataset, the platform will integrate descriptive, diagnostic, predictive, and prescriptive analytics to help identify high-value customers, understand purchasing behavior, predict customer churn, estimate customer lifetime value, and recommend targeted retention strategies.

Rather than focusing solely on predictive modeling, the project adopts a business-first approach in which every analytical component is designed to support a specific organizational decision. The final deliverable will consist of an integrated analytics framework supported by SQL-based data engineering, Python analytics workflows, interactive Tableau dashboards, and a consulting-style executive report.

2. Business Context

2.1 Industry Overview

The global e-commerce industry has experienced sustained growth due to increasing internet penetration, digital payment adoption, and changing consumer preferences. Marketplace platforms such as Amazon, Flipkart, Mercado Livre, and Olist compete not only through product availability but also through customer experience, delivery reliability, pricing strategies, and personalized engagement.

As competition intensifies, organizations face several strategic challenges:

- Rising customer acquisition costs
- Increasing competition for customer loyalty
- Growing demand for personalized experiences
- Pressure to optimize marketing expenditure
- Need for operational efficiency
- Increasing importance of customer retention over acquisition

Research consistently shows that retaining existing customers is substantially more cost-effective than acquiring new ones. Consequently, organizations increasingly rely on analytics to maximize customer lifetime value and improve marketing efficiency.

2.2 Organizational Background

Olist operates as a marketplace connecting thousands of independent sellers with customers across Brazil.

The company collects data across multiple business functions, including:

- Customer registrations
- Orders
- Payments
- Product information
- Seller information
- Delivery performance
- Customer reviews
- Geographic information

Although these datasets provide a comprehensive view of customer interactions, management currently lacks an integrated analytical framework capable of transforming raw operational data into strategic business insights.

3. Business Problem Statement

Olist possesses extensive customer and transaction data but lacks a structured analytical framework to answer critical business questions regarding customer retention, customer value, marketing effectiveness, and operational performance.

Current decision-making is primarily descriptive rather than predictive or prescriptive, limiting the organization's ability to proactively retain valuable customers and optimize marketing investments.

Without customer intelligence, the organization risks:

- Inefficient marketing expenditure
- Declining customer loyalty
- Reduced repeat purchases
- Lower customer lifetime value
- Missed revenue opportunities
- Poor allocation of operational resources

The organization therefore requires an integrated Customer Analytics Platform capable of supporting strategic decision-making across multiple business functions.

4. Business Need

The business requires an analytics platform capable of answering the following strategic questions:

- Which customers generate the greatest long-term value?
- Which customers are likely to discontinue purchasing?
- Which customer segments should receive retention campaigns?
- What operational factors reduce customer satisfaction?
- Which product categories encourage repeat purchases?
- How should marketing resources be allocated to maximize return on investment?

5. Project Objectives

The project seeks to achieve the following objectives.

1. Develop a unified customer view by integrating transactional, behavioral, geographic, and operational data.
2. Identify meaningful customer segments based on purchasing behavior and customer value.
3. Develop a predictive framework for identifying customers at risk of churn.
4. Estimate Customer Lifetime Value (CLV) to support differentiated marketing strategies.
5. Evaluate the impact of operational factors such as delivery performance and customer reviews on customer retention.

6. Provide actionable recommendations that improve customer retention, marketing effectiveness, and revenue growth.

6. Stakeholder Analysis

Stakeholder	Business Objective	Decisions Supported
CEO	Sustainable revenue growth	Strategic planning
Chief Marketing Officer	Improve marketing ROI	Customer targeting
CRM Team	Improve customer retention	Retention campaigns
Operations Team	Improve delivery performance	Logistics optimization
Product Team	Optimize product assortment	Category strategy
Finance Team	Improve profitability	Budget allocation

7. Business Scope

In Scope

- Customer analytics
- Customer segmentation
- Customer retention analysis
- Customer Lifetime Value estimation
- Churn prediction
- Product category analysis
- Geographic analysis
- Delivery performance analysis
- Customer satisfaction analysis
- Executive dashboards
- Business recommendations

Out of Scope

- Fraud detection
- Dynamic pricing
- Inventory optimization
- Supply chain optimization
- Seller recommendation systems
- Real-time analytics
- Production deployment

8. Assumptions

The project assumes that:

- Historical customer behavior reflects future purchasing tendencies.
- Customer identifiers accurately represent unique customers.
- Transaction records are complete and reliable.
- Customer reviews reasonably reflect customer satisfaction.
- Delivery timestamps accurately capture operational performance.
- The available data is representative of business operations during the study period.

9. Risks and Constraints

Data Risks

- Missing customer information
- Incomplete delivery timestamps
- Sparse repeat purchases
- Class imbalance in churn prediction
- Data quality inconsistencies

Business Risks

- Dataset may not contain campaign history
- Marketing costs are unavailable
- Customer demographics are limited
- No direct measure of acquisition cost

Technical Constraints

- Public dataset limitations
- Limited external variables
- No real-time data
- No customer service interactions

10. Success Metrics

Project success will be evaluated using both business and analytical metrics.

Business Success

- Identification of actionable customer segments
- Clear customer personas
- Prioritized retention opportunities
- Identification of revenue concentration
- Executive-ready recommendations
- Interactive decision-support dashboard

Analytical Success

- Reliable churn prediction
- Robust customer segmentation
- Explainable customer lifetime value estimation
- High-quality feature engineering
- Reproducible analytical workflow

11. Business Questions

Customer Value

- Who are the highest-value customers?
- How concentrated is revenue among customers?
- What characteristics define high-value customers?

Customer Loyalty

- Which customers exhibit signs of churn?
- Which factors influence repeat purchasing?
- How does purchase frequency evolve over time?

Customer Satisfaction

- Does customer satisfaction influence future purchases?
- Which operational failures reduce customer satisfaction?
- How important are review ratings in customer retention?

Operations

- Do delivery delays affect repeat purchases?
- Which regions experience operational inefficiencies?
- Which logistics indicators should management monitor?

Product Strategy

- Which product categories generate loyal customers?
- Which categories have the highest customer lifetime value?
- Which categories generate mostly one-time purchases?

Geographic Analysis

- Which cities generate the highest revenue?
- Which regions generate the highest customer lifetime value?
- Are operational issues geographically concentrated?

Marketing

- Which customers should receive retention campaigns?
- Which customers deserve premium loyalty benefits?
- Which customer segments provide the highest expected return on marketing investment?

12. Business Hypotheses

The project will test the following hypotheses.

H1 A small proportion of customers contributes a disproportionately large share of total revenue.

H2 Longer delivery delays reduce customer satisfaction.

H3 Lower customer satisfaction decreases the probability of repeat purchasing.

H4 Customer purchasing behavior naturally forms distinct behavioral segments.

H5 Customer Lifetime Value differs significantly across customer segments.

H6 Behavioral variables such as recency, frequency, and monetary value are significant predictors of customer churn.

H7 Certain product categories generate substantially higher customer loyalty than others.

13. Functional Requirements

The platform shall provide:

- Customer segmentation module
- Customer health scoring
- Churn prediction module
- Customer lifetime value estimation
- Geographic performance analysis
- Product category performance analysis
- Executive dashboard
- KPI monitoring
- Customer profiling
- Business recommendation engine

14. Expected Deliverables

The project will produce:

- Business Requirements Document
- Data Understanding Report
- Data Quality Assessment Report
- SQL Data Engineering Scripts
- Python Analytics Pipeline
- Feature Engineering Documentation
- Customer Segmentation Analysis
- Churn Prediction Model
- Customer Lifetime Value Model
- Tableau Executive Dashboard
- Executive Business Report
- Technical Documentation
- GitHub Repository

15. Project Methodology

The project will follow the CRISP-DM framework, adapted to emphasize business decision-making.

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Exploratory Business Analysis
5. Feature Engineering
6. Customer Intelligence Modeling
7. Model Evaluation
8. Business Recommendations
9. Executive Dashboard Development
10. Project Documentation

16. Expected Business Value

Successful implementation of the proposed Customer Analytics Platform would enable the organization to:

- Improve customer retention through targeted interventions.
- Increase marketing efficiency by prioritizing high-value customers.
- Enhance customer experience through operational improvements.
- Support data-driven strategic planning.
- Increase customer lifetime value.
- Optimize allocation of marketing and operational resources.
- Establish a repeatable analytical framework for future customer analytics initiatives.

Ultimately, the project seeks to demonstrate how integrated analytics can transform transactional data into strategic business intelligence that supports sustainable growth and long-term competitive advantage.